Prabhavit Gupta

AI Engineer | Agentic Workflows, Retrieval Pipelines, Model Context Protocol (MCP) & Evals

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Executive Summary

- **Production AI Engineering & Agentic Workflows:** Software engineer with 3 years commercial experience designing, implementing, and shipping production-grade AI systems and agentic workflows for client projects. Hands-on experience developing multi-step agents, tool-using LLM systems, and robust orchestration layers using LangGraph and Python.
- **Model Context Protocol (MCP), Tool Use & Retrieval Pipelines:** Practical builder of standardized agent tool-calling interfaces and custom Model Context Protocol (MCP Server & Client) integrations. Extensive experience architecting high-throughput Retrieval-Augmented Generation (RAG) pipelines over unstructured enterprise data with PostgreSQL pgvector and Qdrant.
- **Client-Facing Engineering, Evals & Observability:** Skilled in client conversations from initial discovery through production delivery. Experienced establishing rigorous evaluation harnesses (measuring precision, recall, and context faithfulness with RAGAS), token cost/latency optimizations, structured logging, and robust CI/CD pipelines.

Technical Skills

- **Agentic Systems & Frameworks:** LangGraph, LangChain, Multi-Agent Systems, ReAct Planning, Tool Calling, Function Invocations, State Machines, Fallbacks, Human-in-the-Loop Escalation.
- **Model Context Protocol (MCP) & LLMs:** MCP Server & Client Development, Claude 3.5 Sonnet, OpenAI GPT-4o, Prompt Engineering, Structured Outputs (Pydantic / JSON Schema), Guardrails against Hallucinations.
- **Retrieval Pipelines & Vector Stores:** Retrieval-Augmented Generation (RAG), Semantic Chunking, Vector Embeddings, PostgreSQL pgvector, Qdrant, ChromaDB, Hybrid Search (Dense + BM25), Semantic Caching (Redis).
- **Core Software Engineering:** Python (Advanced - Packaging, AsyncIO, PyTest, Type Hints), TypeScript, Node.js, REST/gRPC APIs, Docker, Git, CI/CD Pipelines (GitHub Actions), Microservices Architecture.
- **Evaluation, Observability & Safety:** RAGAS Evaluation Framework, Precision/Recall/F1 Benchmarking, OpenTelemetry Distributed Tracing, Structured Logging, Token Budget Optimization, AI Safety.

Professional Experience

AI Engineer & Solutions Consultant • *Independent / AI Product Engineering Consultancy*
2024 – Present • Venice, IT (Remote Europe)

- Designed, built, and shipped production-grade AI systems, agentic workflows, and retrieval pipelines for European enterprise clients and scale-ups.
- Engineered custom Model Context Protocol (MCP) connectors, allowing autonomous AI agents to securely query relational databases, internal SaaS tools, and multi-repository codebases.
- Architected end-to-end RAG pipelines utilizing semantic chunking, dense vector embeddings, and Cohere reranking on PostgreSQL pgvector, achieving sub-second latency and zero hallucination on source documents.
- Built automated evaluation pipelines and regression test harnesses using RAGAS, establishing systematic benchmarks for model performance, faithfulness, and response relevance across version updates.
- Packaged AI services into clean, containerized Docker microservices and deployed automated CI/CD pipelines with structured logging and OpenTelemetry tracing for real-time observability.

Full-Stack AI Software Engineer • *Tech Automation Systems*
2023 – 2024 • Milan / Remote, IT

- Built asynchronous microservices in Python and TypeScript for high-volume document ingestion, extraction, and structured classification from unstructured PDFs.
- Implemented vector indexing workflows and background queuing systems with Redis and PostgreSQL, ensuring scalable and resilient data processing.
- Created robust unit and integration test suites using PyTest, maintaining high code quality and test coverage across the deployment lifecycle.
- Participated directly in client technical sessions, translating ambiguous operational challenges into validated software architectures and actionable roadmaps.

Key Engineering Projects

- **Autonomous Tool-Using Agent Platform (LangGraph + Python + MCP):** Multi-agent collaborative system executing verified tool operations via Model Context Protocol against enterprise databases, featuring state preservation and human approval gates.
- **Production Document Retrieval Engine (Python + pgvector):** End-to-end knowledge engine combining

dense vector search, hybrid keyword retrieval, semantic caching, and automated hallucination guardrails over complex institutional records.

Education, Languages & Legal Status

- **Bachelor of Arts in Philosophy & Economics** – Ca' Foscari University of Venice (2018 – 2022)
  Rigorous education in formal logic, algorithmic decision theory, analytical reasoning, and complex systems modeling.
- **Languages:** English (C1 Professional, fluent written/spoken), Italian (Working proficiency).
- **Work Authorization & Legal Status:** EU Long-Term Resident (Permesso di Soggiorno UE a Tempo Indeterminato, Italy) – Unrestricted legal right to live and work across the European Union without visa sponsorship; available immediately for remote engagement.